

Choose Parameter Values

$r = 1$ for bias, variance

$r = 2$ for distribution, quantiles

$$c_2 = \begin{cases} 1; & \text{for } r = 1 \\ 0.1; & \text{otherwise} \end{cases}$$

Calculate Bootstrap Estimates

(1) Choose initial block size, $\ell = \frac{1}{n^{r+4}}$, where n is the sample size

(2) Calculate MBB estimators, $\hat{\varphi}_n(\ell)$ and $\hat{\varphi}_n(2\ell)$

Estimate $\widehat{\text{VAR}}$

(1) Set $m = c_2 n^{1/3} \ell^{2/3}$ (no. of blocks to be deleted)

(2) Calculate the Jackknife-after-Bootstrap (JAB) Variance estimator $\widehat{\text{VAR}}$ using $\hat{\varphi}_n(\ell)$

(i) Collect the indices set:

$$I_i^* = \{k: 1 \leq k \leq K, \{ {}_k B_1^*, \dots, {}_k B_b^* \} \cap \{B_i, \dots, B_{i+m-1}\} = \emptyset\}$$

(ii) $\hat{\varphi}_n^{(i)}$: i -th block-deleted jackknife point value, calculated using $k \in I_i^*$

$$(iii) \tilde{\varphi}_n^{(i)} = m^{-1} \left[N \hat{\varphi}_n - (N - m) \hat{\varphi}_n^{(i)} \right]$$

(iv) $\widehat{\text{VAR}}_{\text{JAB}}[\hat{\varphi}_n(\ell)] = \frac{m}{N-m} \frac{1}{M} \sum_{i=1}^M (\tilde{\varphi}_n^{(i)} - \hat{\varphi}_n)^2$, where, N = No. of all possible blocks, and $M = N - m + 1$

Calculate $\hat{C}_1$ and $\hat{C}_2$

$$\hat{C}_1 = n \ell^{-r} \widehat{\text{VAR}}_{\text{JAB}}[\hat{\varphi}_n(\ell)]$$

$$\hat{C}_2 = 2\ell \{ \hat{\varphi}_n(\ell) - \hat{\varphi}_n(2\ell) \}$$

Calculate Optimal Block Length, $\hat{\ell}^0$

$$\hat{\ell}^0 = \left[\frac{2\hat{C}_2^2}{r\hat{C}_1} \right]^{\frac{1}{r+2}} \frac{1}{n^{r+2}}$$